

Simulation Revisit

Jianhui Gao

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Data Generation

To generate $(X^T, W^T, A)^T$, we first simulated 16 random variables from a zero-mean multivariate normal distribution with a first-order autoregressive covariance structure. The covariance between the k -th and l -th random variables was set to $3(0.4)^{|k-l|}$. The first 10 random variables were used as the features, X , for model training, and the next 5 as auxiliary covariates, W , for validation. The final random variable was used to obtain the protected attribute, A , and set to 1 if the normally generated value exceeded 0.253 and set to 0 otherwise. The outcome Y was generated under two different scenarios to induce varying degrees of misspecification into the ML and imputation models:

- Scenario 1 (Old): Logistic activation function with non-linear inputs

$$P(Y = 1|X, W, A = a) = \text{expit}\{-2.3 + \beta_a^T X + \gamma_a^T W + 0.2X_2^2 - 0.1X_3^2 + 0.1X_5X_6\}$$

- Scenario 2 (Old): Non-logistic activation function with non-linear hidden layer

$$P(Y = 1|X, W, A = a) = \exp\{-(1.3 + \beta_a^T X + \gamma_a^T W)^2\}$$

- Scenario 3 (New): Tree Structure

$\eta = \beta^T X + \gamma^T + X_4^2 - X_5^2 + 0.3X_1W_1 - 0.3X_2W_2 + 0.2X_3W_3 + \tanh(0.4W_5) - \tanh(0.2X_4) + 0.1\sin(2W_4X_5)$
and $\psi = \beta^T X + \gamma^T - 0.4W_4^2 + 0.3W_5^2 + 0.2X_1W_1 - 0.2X_2W_2 + 0.1X_3W_3 + \tanh(0.3X_5) - \tanh(0.1W_4) + 0.1\cos(2X_4W_5)$. Then generate S as:

$$S = \begin{cases} 0.8 & \text{if } A = 0, \eta_a > 0.5, \psi_a > 0.3 \\ 0.6 & \text{if } A = 0, \eta_a > 0.5, \psi_a \leq 0.3 \\ 0.4 & \text{if } A = 0, \eta_a \leq 0.5, \psi_a > 0.3 \\ 0.2 & \text{if } A = 0, \eta_a \leq 0.5, \psi_a \leq 0.3 \\ 0.7 & \text{if } A = 1, \eta_a > 0.3, \psi_a > 0.5 \\ 0.5 & \text{if } A = 1, \eta_a > 0.3, \psi_a \leq 0.5 \\ 0.3 & \text{if } A = 1, \eta_a \leq 0.3, \psi_a > 0.5 \\ 0.1 & \text{if } A = 1, \eta_a \leq 0.3, \psi_a \leq 0.5 \end{cases}$$

Finally $P(Y = 1 | X, W, A = a) = S$

- Scenario 4 (New): Log-log activation function with non-monotone effect

$$P(Y = 1|X, W, A = a) = 1 - \exp\{-\exp(g_a(Z))\}$$

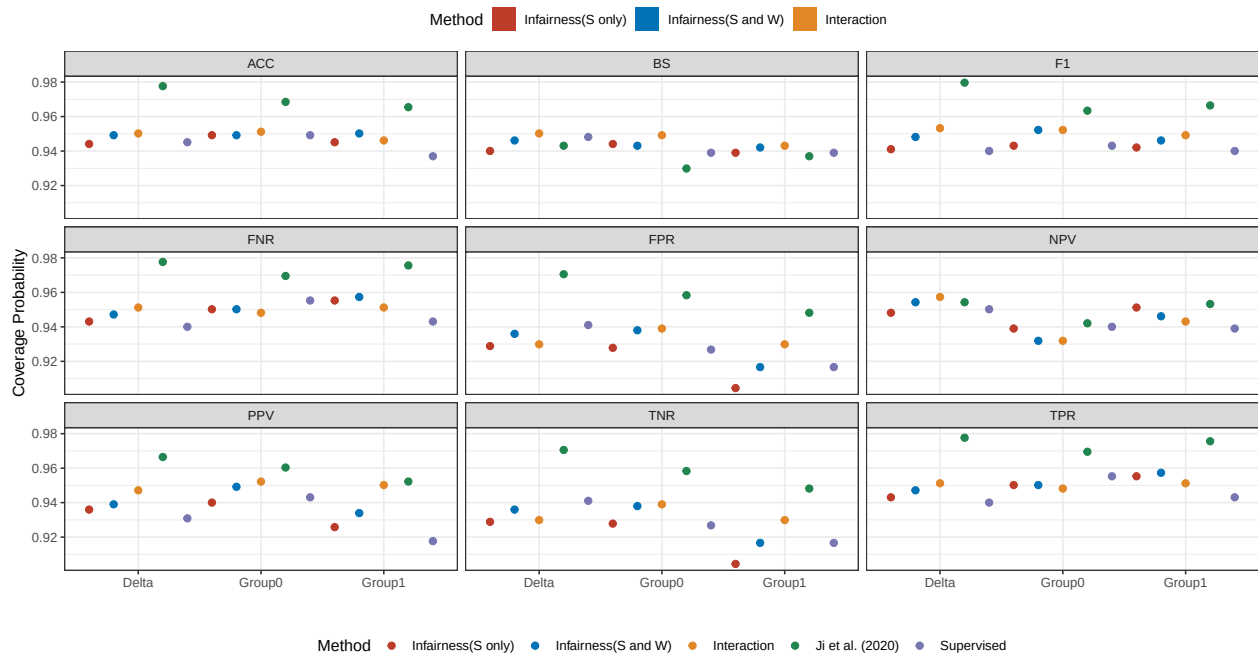
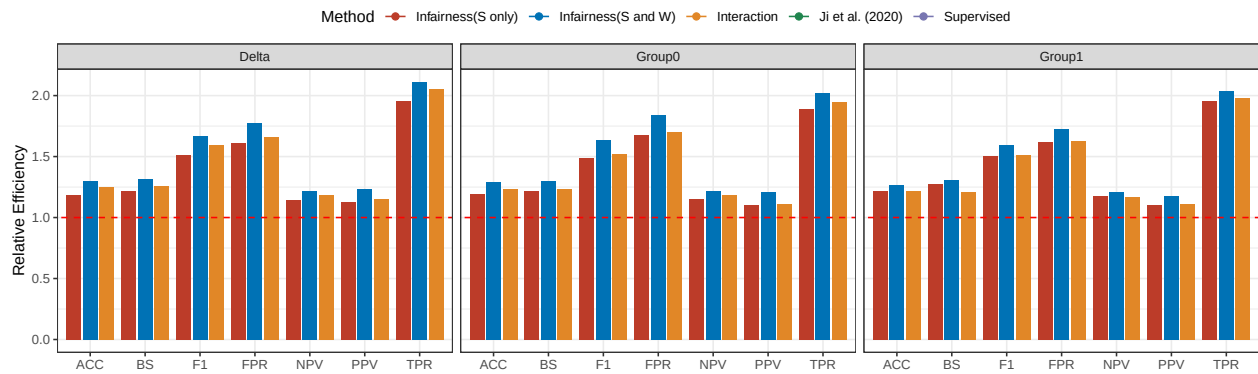
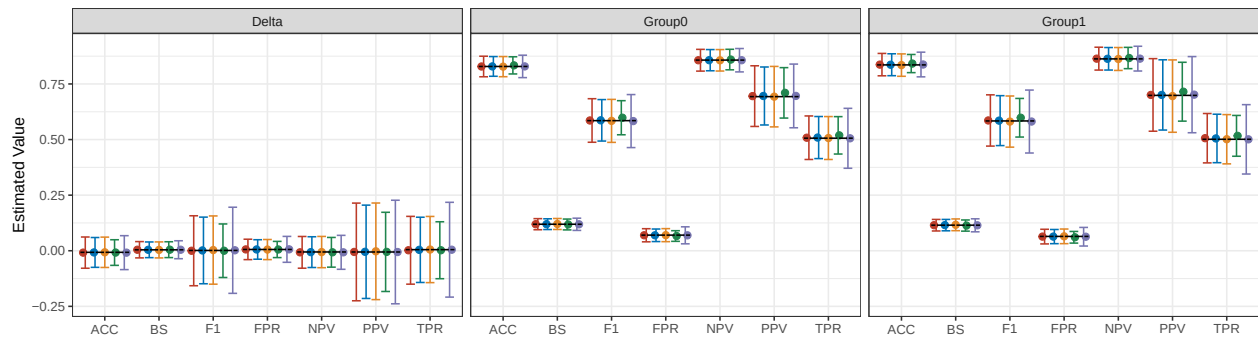
where

$$Z = -0.5 + \beta_a^T X + \gamma_a^T W + 0.1X_1W_1 - 0.1X_2X_5 + 0.12X_3^2 - 0.08X_3^3/3 + 0.10W_1^2$$

and

$$g_a(z) = b_a z + u_{1a} \exp\left\{-\frac{(z+0.30)^2}{2s_0^2}\right\} + d_a \exp\left\{-\frac{z^2}{2s_0^2}\right\} + u_{2a} \exp\left\{-\frac{(z-0.30)^2}{2s_0^2}\right\} + r_a \exp\left\{-(z/0.50)^2\right\} \sin(2\pi k \sigma(z))$$

Scenario 1



##	Method	Group	Brier_Score	Log_Loss	AUC	ECE
## 1	ss_s	g0	0.1220663	0.3861919	0.8516296	0.05574710
## 2	ss_s	g1	0.1180538	0.3754241	0.8523874	0.06005133
## 3	ss_sw	g0	0.1152729	0.3685237	0.8667030	0.05251269
## 4	ss_sw	g1	0.1169644	0.3758470	0.8527594	0.06073705
## 5	ss_sw_int	g0	0.1202156	0.3884099	0.8565405	0.05848497
## 6	ss_sw_int	g1	0.1235647	0.4045494	0.8386574	0.06849891

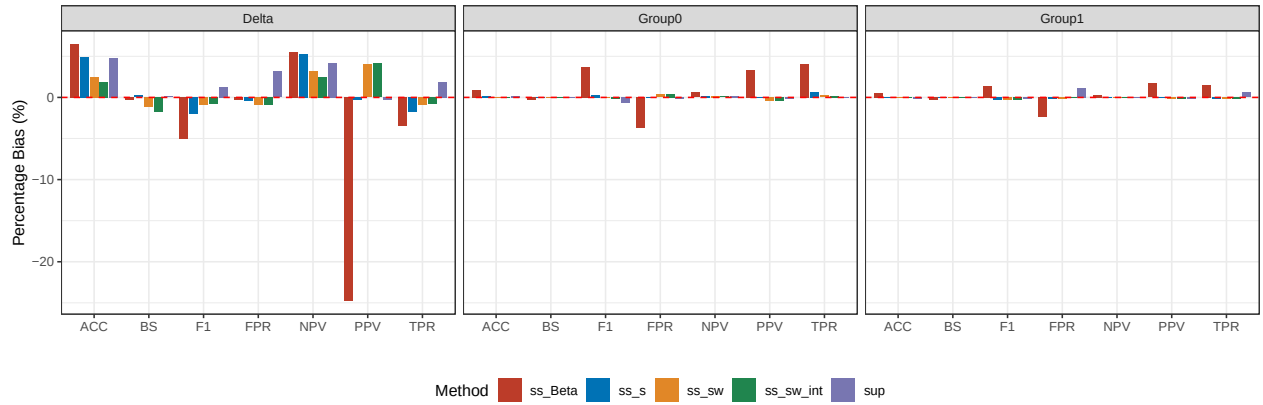
Scenario 2



##	Method	Group	Brier_Score	Log_Loss	AUC	ECE
## 1	ss_s	g0	0.1583596	0.4833568	0.8172537	0.06440370
## 2	ss_s	g1	0.1475951	0.4561913	0.8302624	0.06812091
## 3	ss_sw	g0	0.1243053	0.4095569	0.8879697	0.05619028
## 4	ss_sw	g1	0.1240422	0.4021592	0.8793991	0.05851992
## 5	ss_sw_int	g0	0.1302927	0.4277376	0.8783765	0.05795497
## 6	ss_sw_int	g1	0.1305801	0.4296554	0.8686829	0.06547316

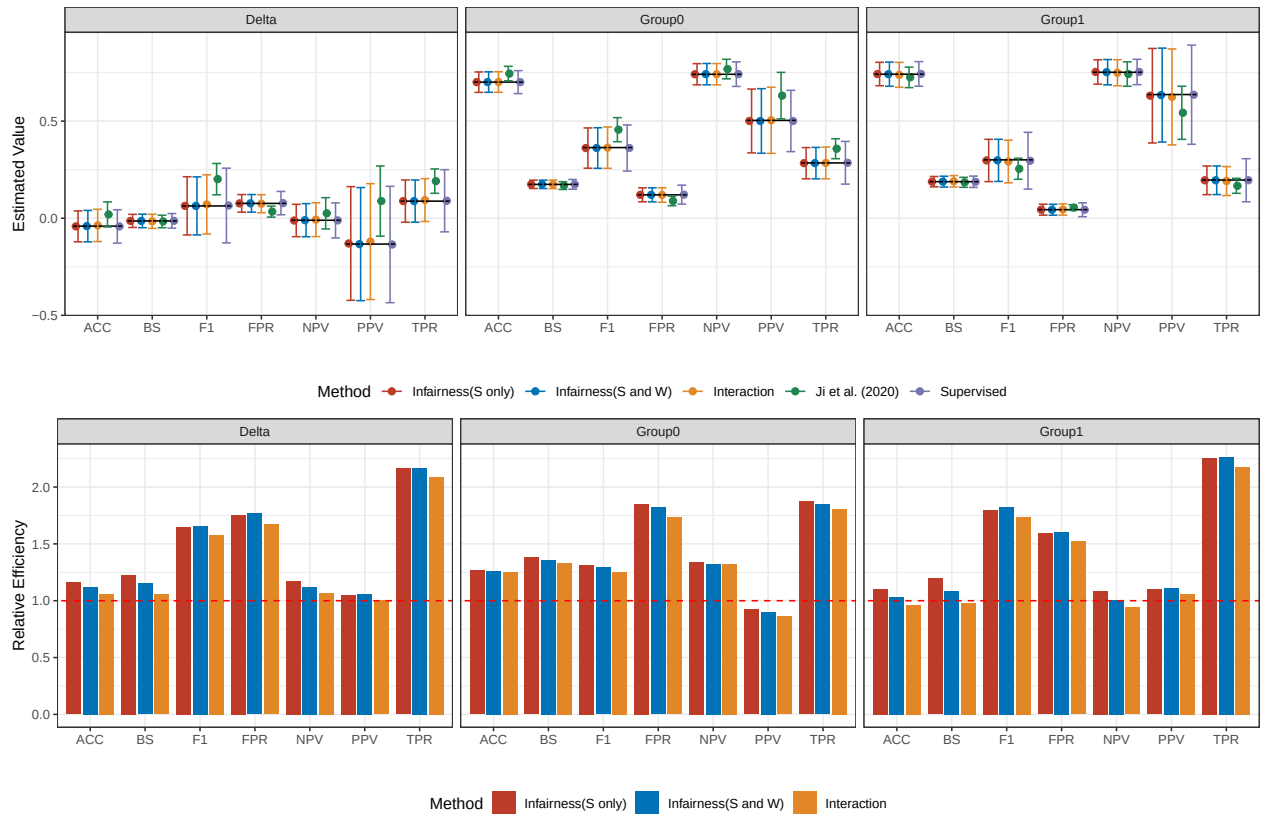
Scenario 3 (Tree Model)

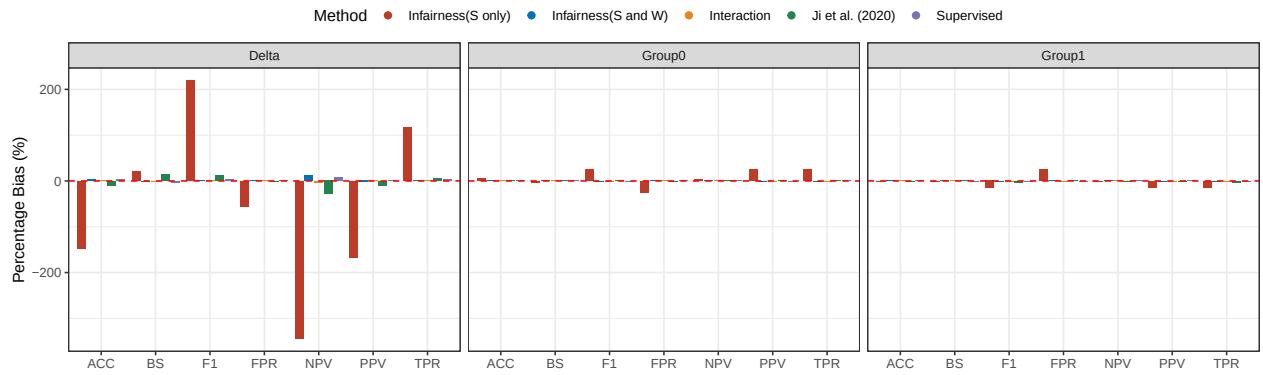
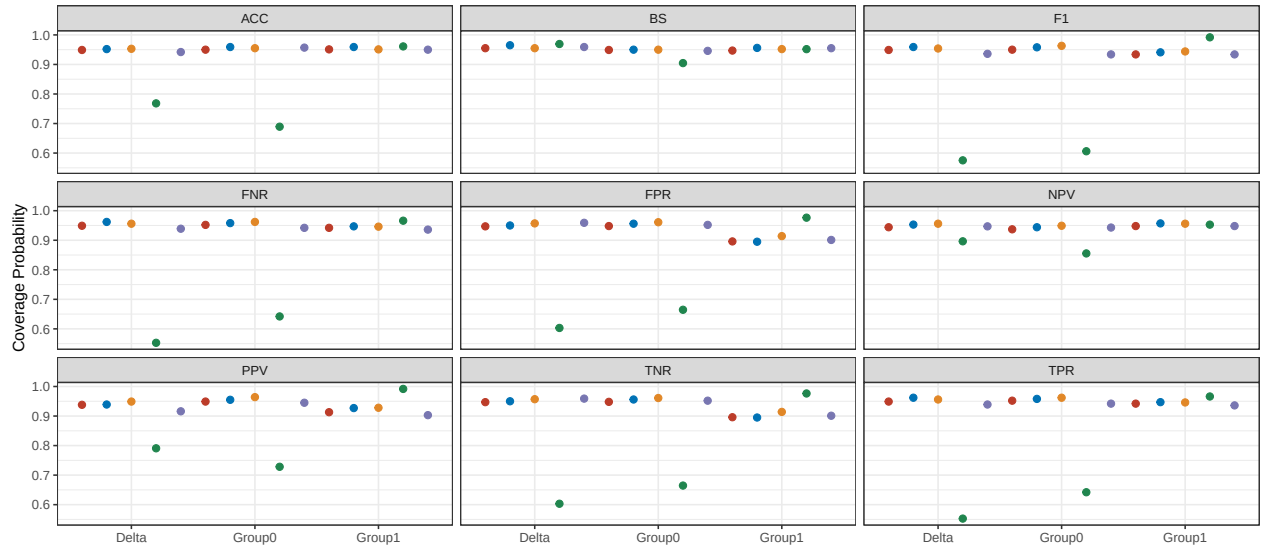




##	Method	Group	Brier_Score	Log_Loss	AUC	ECE
## 1	ss_s	g0	0.2246807	0.6413522	0.6046622	0.07928543
## 2	ss_s	g1	0.2360892	0.6654231	0.5956970	0.09437094
## 3	ss_sw	g0	0.2081914	0.6081860	0.6922573	0.07754860
## 4	ss_sw	g1	0.2219570	0.6396471	0.6774624	0.09307664
## 5	ss_sw_int	g0	0.2112836	0.6200554	0.6859284	0.08536390
## 6	ss_sw_int	g1	0.2257625	0.6535728	0.6705882	0.10040694

Scenario 4





##	Method	Group	Brier_Score	Log_Loss	AUC	ECE
## 1	ss_s	g0	0.1666531	0.4960169	0.7837806	0.07315049
## 2	ss_s	g1	0.1704622	0.5346916	0.7324662	0.09745524
## 3	ss_sw	g0	0.1738822	0.5184140	0.7725569	0.08061780
## 4	ss_sw	g1	0.1944425	0.5779367	0.6112187	0.10315380
## 5	ss_sw_int	g0	0.1779614	0.5375839	0.7639289	0.08336252
## 6	ss_sw_int	g1	0.2011272	0.5967126	0.5922630	0.11685659